1 Power Series

Definition 1.1. A *power series centered at a* is a series of the form

$$\sum_{k=0}^{\infty} c_k(x-a)^k = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots, \tag{1.2}$$

where x is a variable, a is a fixed real number called the *center* of the series, and the c_k's are constants called the *coefficients* of the series.

Remark 1.3. (a) Note that each partial sum $\sum_{k=0}^{n} c_k(x-a)^k$ is a polynomial in $(x-a)$.

(b) In (1.2), we have adopted the convention that $(x-a)^0 = 1$, even when $x = a$.

(c) When $a = 0$, (1.2) becomes $\sum_{k=0}^{\infty} c_k x^k = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \cdots$.

For each fixed x, the series (1.2) is a series of constants that we can test for convergence or divergence. A power series may converge for some values of x and diverge for other values of x.

1. For a fixed collection of coefficients $\{c_n\}$ and center a, define a function f by $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$. What is the domain of f?

2. Show that every power series converges at $x = a$. What value does it converge to?

3. If we take $a = 0$ and $c_n = 1$ for all n, then (1.2) becomes $\sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \cdots$. For what values of x does this converge? What value does it converge to?

4. For what values of x does the series $\sum_{k=0}^{\infty} k!x^k$ converge? (*Hint:* Use the Ratio Test.)

5. For what values of x does the series $\sum_{k=1}^{\infty} \frac{(x-3)^k}{k}$ converge? (*Hint:* Use the Ratio Test.)

6. The *Bessel function* $J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{2^{2k}(k!)^2}$ has important applications in physics. What is the domain of $J_0(x)$?

2 Interval of Convergence

For the power series we have looked at so far, the set of values of x for which the series converges has always turned out to be an interval, or a single point. The following theorem says this is true in general.

Theorem 2.1. *For a given power series $\sum_{k=0}^{\infty} c_k(x-a)^k$ there are only three possibilities:*

(i) The series converges only when $x = a$.

(ii) The series converges for all x.

(iii) There is a positive number R such that the series converges if $|x-a| < R$ and diverges if $|x-a| > R$.

Definition 2.2. The number R in case (iii) above is called the *radius of convergence* of the power series. By convention, we define $R = 0$ in case (i) and $R = \infty$ in case (ii).

We can write $|x-a| < R$ as $a - R < x < a + R$. When x is an endpoint of the interval $[a-R, a+R]$, that is $x = a \pm R$, anything can happen - the series might converge at one or both endpoints or it might diverge at both endpoints. Thus in case (iii) there are four possibilities for the *interval of convergence*:

$$(a-R, a+R) \quad (a-R, a+R] \quad [a-R, a+R) \quad [a-R, a+R].$$

7. For each series below, use the ratio test to determine the radius and interval of convergence.

(a) $\displaystyle\sum_{k=1}^{\infty} \frac{x^k}{k!}$

(b) $1+\frac{2x}{2^3}+\frac{4x^2}{3^3}+\frac{8x^3}{4^3}+\frac{16x^4}{5^3}+\frac{32x^5}{6^3}+\cdots$.

(c) $\sum_{k=1}^{\infty}\frac{(x-2)^k}{k^2 3^k}$